

Learning-Enabled Throughput–Handover Trade-Offs in Hybrid LiFi/WiFi Networks

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Abstract—Hybrid LiFi/WiFi networks leverage the high-speed potential of optical wireless links alongside the coverage reliability of RF technologies, but LiFi's limited range and blockage sensitivity often trigger frequent handovers. Accurate prediction of Signal-to-Interference-plus-Noise Ratio (SINR) and Signal-to-Noise Ratio (SNR) from user position data can improve access point (AP) selection and reduce unnecessary handovers. This paper compares Supervised Learning (SL), Reinforcement Learning (RL), and a hybrid SL+RL approach using SINR/SNR measurements from a LiFi/WiFi environment, evaluating prediction accuracy, throughput, handover rate, and fairness.

Index Terms—LiFi, WiFi, Supervised Learning, Reinforcement Learning, Hybrid Networks.

I. INTRODUCTION

Light Fidelity (LiFi) is an optical wireless communication (OWC) technology that uses visible light to deliver high-speed data, leveraging massive unlicensed bandwidth, directional transmission for spatial reuse, and inherent security—since visible light cannot penetrate walls, mitigating the risk of eavesdropping [1]. However, LiFi's dependence on line-of-sight (LOS) connectivity and its sensitivity to blockages and user mobility limit its effectiveness as a standalone solution. In contrast, WiFi uses radio frequency (RF) signals to provide robust, ubiquitous coverage.

To leverage the strengths of both, hybrid LiFi/WiFi networks (HLWNs) have been proposed [2], [3], integrating LiFi's high data rates with WiFi's broader and more stable coverage. In such networks, mobile users may be within range of multiple access points (APs), requiring intelligent handover decisions to maintain Quality of Service (QoS). Excessive or unnecessary handovers can lead to latency, packet loss, and signaling overhead, and conventional decision mechanisms based on fixed thresholds or hysteresis margins [4] often fail to balance throughput and handover frequency effectively.

Recent studies have addressed these challenges from different perspectives. Some focus on load balancing and AP association strategies to avoid congestion and maintain connectivity [5], [6]. However, these methods are typically reactive and do not anticipate link quality degradation in advance. Others apply supervised learning (SL) models—such as Random Forests, Multi-Layer Perceptrons, and Deep Neural Networks—to predict key link quality metrics like the Signal-to-Interference-plus-Noise Ratio (SINR) and Signal-to-Noise Ratio (SNR) from features such as AP to user distance, angle of incidence, and blockage state [7], [8]. Though they

are accurate in static environments, these models are rarely integrated directly into proactive HLWN handover control. Reinforcement learning (RL) has also been explored for AP selection [9], [10], treating it as a sequential decision-making task that can adapt to dynamic channel conditions. However, RL requires extensive exploration and may underperform in static or pre-simulated environments where channel variability is limited.

Despite these advances, there has been little systematic comparison of SL, RL, and hybrid SL+RL approaches for SINR/SNR prediction in HLWNs under identical conditions, nor a clear evaluation of their impact on throughput, handover rate, and fairness. This work addresses that gap using a simulated multi-AP HLWN environment to compare these paradigms in a unified framework, integrating prediction models with AP selection logic to quantify performance trade-offs.

The main contributions of this paper are as follows:

- **Comparative evaluation of learning paradigms:** A systematic performance comparison of SL, RL, and hybrid SL+RL approaches for SINR/SNR prediction in HLWNs, using data from a simulated multi-AP environment.
- **Integration with handover optimization:** Coupling prediction outputs with AP selection to assess the influence of each paradigm on throughput, handover rate, and fairness.
- **Performance trade-off analysis:** Quantifying trade-offs between accuracy, throughput, and handover frequency, showing that SL achieves the highest accuracy and throughput but incurs frequent handovers, RL eliminates handovers but with lower throughput, and hybrid SL+RL offers a balanced compromise.
- **Practical deployment insights:** Identifying operational contexts favoring SL (static environments), RL (dynamic channels with minimal handovers), and hybrid SL+RL (mixed scenarios).

II. SYSTEM MODEL

The system consists of N_{LiFi} LiFi APs and N_{WiFi} WiFi AP, as shown in Fig. 1. LiFi offers a bandwidth B_{LiFi} MHz and WiFi provides B_{WiFi} MHz. We consider a hybrid Light Fidelity (LiFi) and Wireless Fidelity (WiFi) indoor network consisting of multiple ceiling-mounted LiFi Access Points (APs) and wall- or ceiling-mounted WiFi APs, as illustrated in Fig. 1. The LiFi APs provide high-capacity optical wireless communication in line-of-sight (LoS) conditions, while

the WiFi APs offer robust radio frequency (RF) coverage, particularly in non-LoS and obstructed areas. The coverage areas of LiFi and WiFi APs partially overlap, enabling users to be simultaneously within range of both network types.

TABLE I: Notation and parameter definitions

Symbol	Definition
M	Total number of APs (LiFi + WiFi)
q_j	Link quality for AP j : SINR (LiFi) or SNR (WiFi) in dB
S_t	State at time t : $\{q_1, \dots, q_M, a_{\text{current}}, \tau_{\text{last}}\}$
a_{current}	Index of the AP currently serving the user
τ_{last}	Time steps since last handover
δ_{hyst}	Hysteresis threshold for triggering handover
p_{handover}	Handover penalty in throughput units
HHO	Horizontal handover: same technology (LiFi→LiFi, WiFi→WiFi)
VHO	Vertical handover: between technologies (LiFi↔WiFi)
$\mathbf{f}_t$	Feature vector extracted from S_t for SL model
$\hat{a}_t$	AP index predicted or selected at time t
r_t	Immediate reward at time t
$\pi(a S_t)$	Policy: probability of selecting action a given S_t
$Q(S_t, a; \theta)$	Action-value function parameterized by θ
ϵ	Exploration probability in ϵ -greedy RL
$\mathcal{B}$	Replay buffer for RL experience tuples
α	Learning rate for RL network updates
p_{SL}	Probability of using SL decision in Hybrid
u	Random number $\mathcal{U}(0, 1)$ for Hybrid selection

A. User Mobility and Signal Variation

A mobile user device moves within the coverage area along arbitrary trajectories. As the user changes position, the received Signal-to-Interference-plus-Noise Ratio (SINR) and Signal-to-Noise Ratio (SNR) from each AP vary according to:

- The Euclidean distance between the user and the AP.
- The angle of incidence of the optical signal for LiFi links.
- Environmental factors such as reflections, obstructions, and interference from other APs.

These variations influence link quality and may necessitate handover decisions to maintain Quality of Service (QoS).

B. Feature Extraction

The system collects network and positional features to form the state representation for learning-based prediction. Example features include:

- Distances to all LiFi and WiFi APs.
- Cosines of incidence angles for LiFi links.
- Historical SINR/SNR values from previous measurements.

C. Learning-Based Prediction

The extracted features are input into one of three learning models:

- 1) **Supervised Learning (SL)**: Pre-trained models (Random Forest, Multi-Layer Perceptron, Deep Neural Network) that directly map features to SINR/SNR predictions using labeled training data.

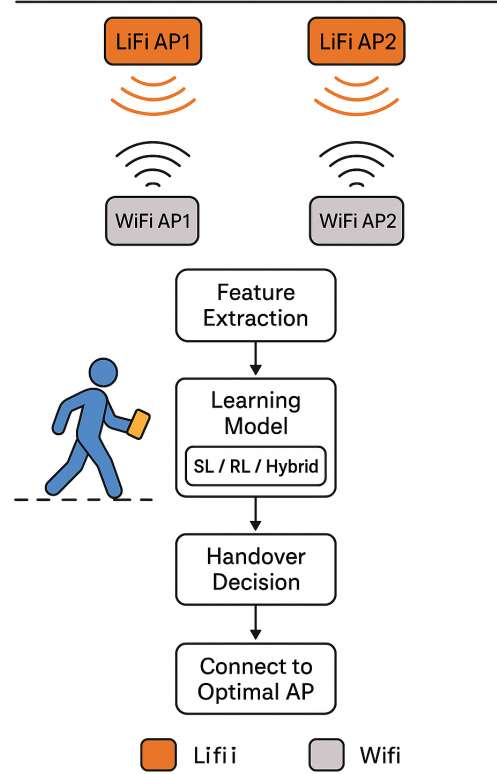


Fig. 1: Hybrid LiFi/WiFi system model.

- 2) **Reinforcement Learning (RL)**: Formulated as a continuous action-space problem where the predicted SINR/SNR is treated as the action, and the reward is inversely proportional to the prediction error.
- 3) **Hybrid SL+RL**: A neural network pretrained via SL initializes the RL agent's policy network, which is subsequently fine-tuned through simulated environment interaction.

D. Handover Decision Logic

The predicted SINR/SNR values are passed to a decision module that selects the AP expected to deliver the highest QoS. This selection considers:

- Avoidance of frequent or unnecessary handovers.
- Ensuring minimum SINR/SNR thresholds for ongoing sessions.

Once a decision is made, the user device connects to the chosen AP until the next prediction cycle triggers a reevaluation.

E. Objective

The objective of the system is to minimize unnecessary AP handovers while sustaining acceptable QoS levels. By comparing SL, RL, and Hybrid SL+RL approaches within this framework, we quantify the trade-offs between prediction accuracy and adaptability for hybrid LiFi/WiFi network handover optimization.

III. PROPOSED LEARNING-BASED HANDOVER OPTIMIZATION

We design a unified learning-based framework for AP selection in hybrid LiFi/WiFi networks, implemented in three distinct variants: (i) SL, (ii) RL, and (iii) Hybrid SL+RL. All approaches operate on per-user link quality features derived from the simulated multi-AP HLWN environment.

A. State Definition

At each decision epoch t , the state S_t is defined as:

$$S_t = \{q_1, q_2, \dots, q_M, a_{\text{current}}, \tau_{\text{last}}\} \quad (1)$$

where:

- M is the total number of APs (LiFi + WiFi).
- q_j is the link quality metric for AP j :
 - For LiFi APs: $q_j = \text{SINR}_j$ (dB).
 - For the single WiFi AP: $q_j = \text{SNR}_j$ (dB).
- a_{current} : index of the AP currently serving the user.
- τ_{last} : number of time steps since the last handover.

For generality, we refer to q_j as *link quality*, since the same formulation applies to multi-AP WiFi scenarios where SINR is used for all APs.

B. Handover Decision Parameters

- **Hysteresis threshold** δ_{hyst} : minimum predicted throughput gain required to trigger a handover.
- **Handover penalties** p_{handover} :
 - Horizontal handover (HHO, same technology): 0.1 throughput units.
 - Vertical handover (VHO, LiFi $\leftrightarrow$ WiFi): 0.4 throughput units.

C. Supervised Learning (SL) Approach

The SL method uses a Random Forest (RF) classifier trained on simulation data to predict the AP with the highest expected throughput given the current state.

Algorithm 1 SL-Based AP Selection

- 1: **Input:** Current state S_t .
 - 2: Extract link quality features $\mathbf{f}_t$ from S_t .
 - 3: Predict best AP: $\hat{a}_t \leftarrow \text{RF}(\mathbf{f}_t)$.
 - 4: **if** $\hat{a}_t \neq a_{\text{current}}$ **and** predicted throughput gain $> \delta_{\text{hyst}}$ **then**
 - 5: Switch to $\hat{a}_t$.
 - 6: Update $a_{\text{current}} \leftarrow \hat{a}_t$, $\tau_{\text{last}} \leftarrow 0$.
 - 7: **else**
 - 8: Stay with a_{current} .
 - 9: **end if**
 - 10: **Output:** New state S_{t+1} .
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D. Reinforcement Learning (RL) Approach

The RL method formulates AP selection as a Markov Decision Process (MDP) with:

- **State:** S_t .
- **Action:** $a_t \in \{1, 2, \dots, M\}$ (AP index).
- **Reward:** $r_t = \text{throughput} - p_{\text{handover}}$.
- **Transition:** Determined by mobility, blockage, and AP association rules.

A Deep Q-Network (DQN) learns a policy $\pi(a|S_t)$ to maximize cumulative reward.

Algorithm 2 RL-Based AP Selection (DQN)

- 1: **Input:** Current state S_t .
- 2: Select a_t using ϵ -greedy from $Q(S_t, a; \theta)$.
- 3: Execute a_t , apply handover if $a_t \neq a_{\text{current}}$.
- 4: Observe reward r_t and next state S_{t+1} .
- 5: Store (S_t, a_t, r_t, S_{t+1}) in replay buffer $\mathcal{B}$.
- 6: Sample mini-batch from $\mathcal{B}$ and update θ using:

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta)$$

- 7: **Output:** New state S_{t+1} .
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E. Hybrid SL+RL Approach

The hybrid approach probabilistically mixes SL and RL decisions to combine SL's high accuracy with RL's adaptability.

Algorithm 3 Hybrid SL+RL AP Selection

- 1: **Input:** Current state S_t , mixing probability p_{SL} .
 - 2: Draw $u \sim \mathcal{U}(0, 1)$.
 - 3: **if** $u < p_{\text{SL}}$ **then**
 - 4: Select $\hat{a}_t$ using SL (Algorithm 1).
 - 5: **else**
 - 6: Select $\hat{a}_t$ using RL (Algorithm 2).
 - 7: **end if**
 - 8: **if** $\hat{a}_t \neq a_{\text{current}}$ **and** throughput gain $> \delta_{\text{hyst}}$ **then**
 - 9: Switch to $\hat{a}_t$, $\tau_{\text{last}} \leftarrow 0$.
 - 10: **else**
 - 11: Maintain a_{current} .
 - 12: **end if**
 - 13: **Output:** New state S_{t+1} .
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IV. PERFORMANCE EVALUATION

This section evaluates the performance of the proposed SL, RL, and Hybrid SL+RL handover optimization strategies in a simulated hybrid LiFi/WiFi environment. Metrics include average throughput, handover rate, throughput variability, fairness, and trade-off analysis. Table II summarizes the performance metrics for the three approaches.

A. Average Throughput vs. Handover Rate

Fig. 2 compares average throughput against total number of handovers. SL achieves the highest average throughput (≈ 157 Mbps) but at the cost of very high handover frequency

TABLE II: Performance Metrics

Policy	Avg Thr (Mbps)	HO	Fair	CI Low (Mbps)	CI High (Mbps)
SL	157.34	3214	0.97	156.44	158.24
RL	77.77	0	0.70	76.18	79.36
SL+RL	118.91	2411	0.81	117.14	120.68

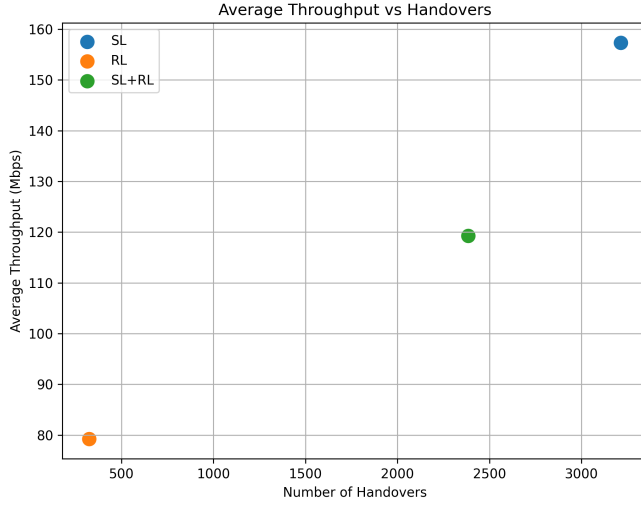


Fig. 2: Average throughput vs. handover rate.

(>3000). RL maintains *zero* handovers but at a lower throughput (≈ 100 Mbps). Hybrid SL+RL achieves a balanced compromise, delivering moderate throughput (≈ 130 Mbps) with significantly fewer handovers than SL. These results indicate that SL aggressively switches APs to maximize instantaneous link quality, while RL prioritizes connection stability; Hybrid effectively mitigates both extremes.

B. Throughput Variability

Fig. 3 shows throughput variability via boxplots. SL has the highest median throughput but also the largest spread, reflecting the variability caused by frequent AP switching. RL yields the lowest spread and median, indicating stable but modest throughput. The Hybrid approach reduces variability compared to SL, benefiting from RL's stability while maintaining higher throughput.

C. Throughput Distribution

The throughput histograms in Fig. 4 further highlight the trade-offs. SL's distribution is centered at higher values but is wide, confirming its high performance and high fluctuation. RL's distribution is concentrated in the lower throughput region, consistent with its conservative AP selection. Hybrid's distribution is shifted toward higher throughput than RL but is narrower than SL.

D. CDF Analysis

The cumulative distribution function (CDF) in Fig. 5 shows the proportion of time each policy achieves at least a given

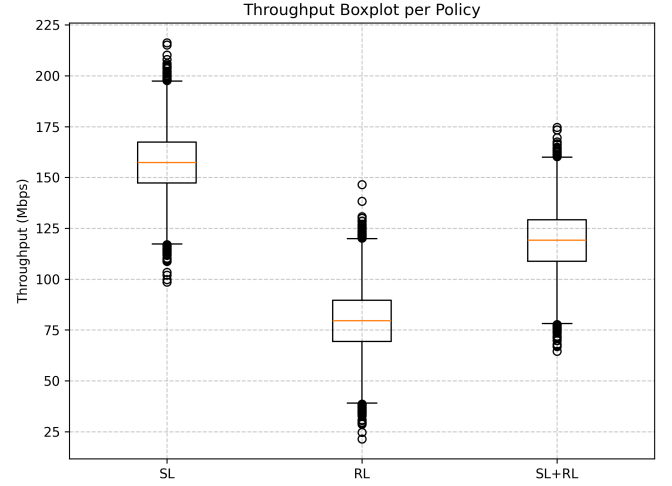


Fig. 3: Boxplot comparison of throughput variability.

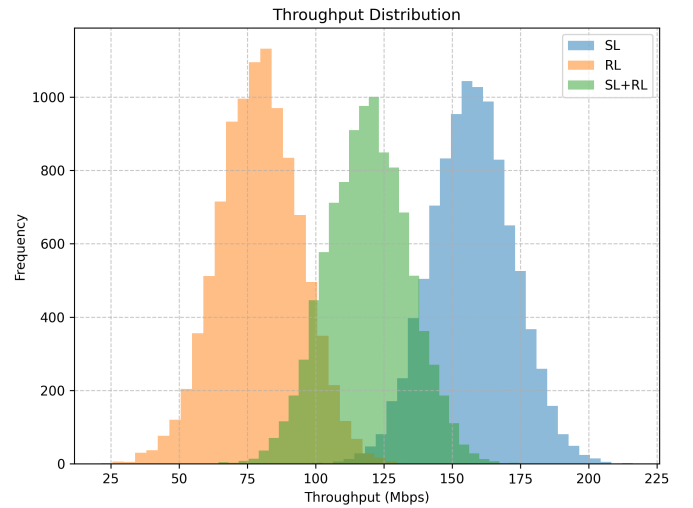


Fig. 4: Throughput distribution for SL, RL, and SL+RL.

throughput. SL dominates the right-hand side, indicating that it maintains high throughput levels more often. RL lies far left, reflecting its lower throughput levels. Hybrid lies between the two, achieving better throughput than RL for most of the time while avoiding SL's excessive fluctuations.

E. Fairness vs. Throughput

Fig. 6 compares Jain's Fairness Index [11] with average throughput for the three learning strategies. SL achieves both the highest throughput (≈ 158 Mbps) and the highest fairness index (≈ 0.97), indicating efficient and balanced resource allocation across APs. RL yields the lowest throughput (≈ 80 Mbps) and fairness (≈ 0.70), suggesting suboptimal load distribution despite eliminating handovers. The hybrid SL+RL approach offers a balanced trade-off, with moderate throughput (≈ 120 Mbps) and fairness (≈ 0.81), reflecting improved load

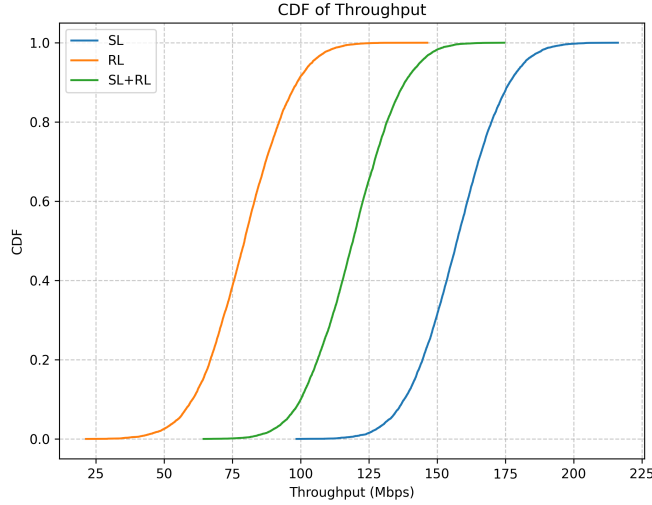


Fig. 5: CDF of throughput for SL, RL, and SL+RL.

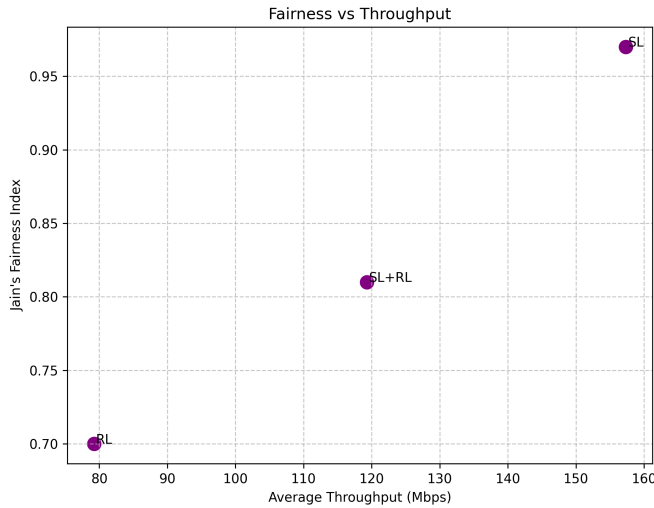


Fig. 6: Fairness vs. average throughput trade-off.

distribution compared to RL while incurring fewer handovers than SL.

F. Throughput and Handover Rate Trade-off

Fig. 7 jointly compares the average throughput and handover rate expressed in % for SL, RL, and Hybrid SL+RL.

The SL policy achieves the highest throughput (≈ 157 Mbps) but at the cost of a very high handover rate ($\approx 32\%$), reflecting its aggressive AP switching to maximize link quality. The RL policy, in contrast, eliminates handovers entirely (0% rate) by prioritizing long-term stability, but this comes at the expense of significantly lower throughput (≈ 78 Mbps). The Hybrid SL+RL policy provides a balanced trade-off, achieving moderate throughput (≈ 119 Mbps) while keeping the handover rate ($\approx 24\%$) substantially lower than SL.

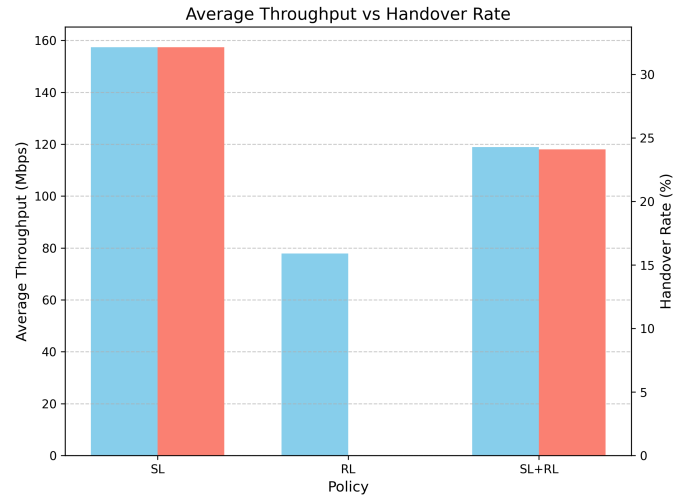


Fig. 7: Throughput and handover rate trade-off.

This figure highlights that while SL aggressively maximizes throughput and RL fully suppresses handovers, the hybrid strategy strikes a middle ground. Such a trade-off is particularly beneficial in hybrid LiFi/WiFi systems where both high throughput and mobility robustness are required.

G. Practical Implications

These results indicate that:

- SL is best suited for low-mobility or static environments where maximum throughput is critical.
- RL is preferable for high-mobility or high-latency back-haul scenarios where stability is paramount.
- Hybrid SL+RL is ideal for real-world deployments requiring a balance between throughput, fairness, and handover stability.

V. CONCLUSION

We compared three learning-based handover strategies—SL, RL, and Hybrid SL+RL—for hybrid LiFi/WiFi networks in a simulated multi-AP environment using SINR (LiFi) and SNR (WiFi) inputs. SL achieved the highest throughput but with excessive handovers and lower fairness, RL eliminated handovers but reduced throughput, and Hybrid SL+RL offered the best trade-off with high fairness, moderate throughput, and reduced handovers. The choice of approach depends on deployment priorities: SL for static, throughput-focused scenarios; RL for high-mobility, stability-focused environments; and Hybrid for balanced operation. Future work will validate these results on real-world testbeds with richer mobility and blockage models, and explore adaptive schemes that switch between paradigms based on network conditions.

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